Levi **Newediuk**

Assistant Professor
Department of Biology, Mount Royal University
403 440 8484 | lnewediuk@mtroyal.ca | ljnewediukecology.com | ljnewediuk

Academic positions

Mount Royal University
Assistant Professor — *2025-Present*

University of Manitoba
Postdoctoral Fellow — *2022-2025*

Education

Memorial University of Newfoundland — *Dr. Eric Vander Wal*
PhD, Biology — *2022*

University of Manitoba — *Dr. James Hare*
MSc, Biology — *2017*

University of Manitoba — *Dr. James Hare*
BSc Hons, Biology — *2011*

Academic profile

Research interests & mentorship

I study how humans shape animal movement. My long-term goal is to learn what drives the movement of individual animals in response to human activity and disturbance, which is becoming an omnipresent feature of our modern landscapes, including those withi and around Calgary. Rather than simply documenting how human disturbance impacts animal movement and distributions, my research aims to determine how individuals integrate information about environmental pressures, resource opportunities, and their physiological states when deciding where to move. I use mostly non-invasive approaches and tools to address my research questions, including behavioural observations to study small-scale movement, camera traps to track movement over larger scales, and extracting physiological data from fecal and hair samples to connect movement to physiological state.

I have mentored the following students:

Sophia Vig-Benko
Research Assistant — *Summer 2026*

Luis Fuenmayor
Research Assistant — *Summer 2026*

Jazmin Singh-Kaur
BIOL 5201 (Project: Temporal changes in raccoon activity) — *Winter 2026*

Teaching experience & interests

I currently teach in the area of ecology and evolution. I am passionate about teaching essential quantitative skills to students, demonstrating through hands-on exercises that math and coding can also be fun. In addition to teaching, I am also developing upper-year courses in population and community ecology. I have taught the following courses:

BIOL 4320: Field Biology Research Techniques
Mount Royal University — *2026*

BIOL 3401: Big Questions & Big Data
Mount Royal University — *2026*

BIOL 2213: Principles of Ecology & Evolution
Mount Royal University — *2025-26*

BIOL 4330: Population & Community Ecology

Mount Royal University *2025*

BIOL 3350: Data Analysis in Ecology

University of Manitoba *2023-2024*

BIOL 1010: Biological Diversity & Interaction

University of Manitoba *2023*

Peer-reviewed publications

In review/revision

1. Whaley, Z. B., Alston, M. A., Bahnson, C., **Newediuk,** L., & Others. (2026). Elk without borders: Genomics reveals asymmetrical gene flow among a patchily-distributed cervid. *Ecological Applications*, *in review*.
2. Peers, M. J., **Newediuk,** L., Boyle, S. P., JG, H., & Others. (2026). Seasonal timing and maternal condition influence the costs of reproduction in a cyclic species. *Journal of Mammalogy*, *in revision*.
3. Funk, K. R., **Newediuk,** L., Brown, L. R., Ganswindt, A., & Vander Wal, E. (2026). Sociality shapes context-dependent stress responses to anthropogenic disturbance in african elephants. *Behavioral Ecology*, *in revision*.

Published

1. **Newediuk,** L., Jesmer, B. R., Mastromonaco, G., & Vander Wal, E. (2026). Landscapes of fearlessness revealed through hormonal responses to putatively risky places. *Behavioral Ecology*, *37*, arag004. https://doi.org/10.1093/beheco/arag004
2. Oliver, R. Y., Yanco, S. W., Others, **Newediuk,** L., & Others. (2026). Interacting effects of human presence and landscape modification on birds and mammals. *Science*, *392*, 879–884. https://doi.org/10.1126/science.adq3396
3. Bath, D., & **Newediuk,** L. (2026). Revisiting size selective mortality in young fish: Do small teleosts really pay a cost? *Canadian Journal of Fisheries and Aquatic Sciences*, *83*, 1–11. https://doi.org/10.1139/cjfas-2025-0212
4. **Newediuk,** L., Richardson, E. S., Bohart, A. M., Roberto-Charron, A., & Others. (2025). Designing epigenetic clocks for wildlife research. *Molecular Ecology Resources*, *25*, e14120. https://doi.org/10.1111/1755-0998.14120
5. Korpach, A. M., de Greef, E., **Newediuk,** L., Schmidt, C., & Others. (2025). Moving to mate? Migration strategy does not predict genetic structure or diversity in bats (chiroptera). *Biological Journal of the Linnean Society*, *145*, blae068. https://doi.org/10.1093/biolinnean/blae068
6. Boyle, S. P., Balsdon, M., **Newediuk,** L., Litzgus, J. D., & Others. (2025). Anuran carcass persistence on roads: Causes and implications for conservation. *The Journal of Wildlife Management*, *89*, e22731. https://doi.org/10.1002/jwmg.22731
7. **Newediuk,** L., Mastromonaco, G. F., & Vander Wal, E. (2024). Associations between glucocorticoids and habitat selection reflect daily and seasonal energy requirements. *Movement Ecology*, *12*. https://doi.org/10.1186/s40462-024-00475-9
8. **Newediuk,** L., & Bath, D. R. (2023). Meta-analysis reveals between-population differences affect the link between glucocorticoids and population health. *Conservation Physiology*, *11*, coad005. https://doi.org/10.1093/conphys/coad005
9. Collins, S. M., Hendrix, J. G., Others, **Newediuk,** L., & Others. (2023). Bibliometric investigation of the integration of animal personality in conservation contexts. *Conservation Biology*, *37*, e14021. https://doi.org/10.1111/cobi.14021
10. **Newediuk,** L., & Vander Wal, E. (2022). Predicting the individual identity of non-invasive faecal and hair samples using biotelemetry clusters. *Mammalian Biology*, *102*, 685–700. https://doi.org/10.1007/s42991-021-00173-8
11. **Newediuk,** L., Prokopenko, C. M., & Vander Wal, E. (2022). Individual differences in habitat selection mediate landscape level predictions of a functional response. *Oecologia*, *198*, 99–110. https://doi.org/10.1007/s00442-021-05098-0
12. Street, G. M., Potts, J. R., Others, **Newediuk,** L., & Others. (2021). Solving the sample size problem for resource selection functions. *Methods in Ecology and Evolution*, *12*, 2421–2431. https://doi.org/10.1111/2041-210X.13701

13. **Newediuk,** L., Ethier, J. P., Boyle, S. P., Aubin, J. A., & Others. (2021). Sociopolitical factors drive conservation planning timelines: A canadian case study with global implications. *Biological Conservation*, *257*, 109091. `https://doi.org/10.1016/j.biocon.2021.109091`
14. **Newediuk,** L. J., & Hare, J. F. (2020). Burrowing richardson's ground squirrels affect plant seedling assemblages via environmental but not seed bank changes. *Current Zoology*, *66*, 219–226. `https://doi.org/10.1093/cz/zoz047`
15. **Newediuk,** L. J., Waters, I., & Hare, J. F. (2015). Aspen parkland pasture altered by richardson's ground squirrel (urocitellus richardsonii sabine) activity: The good, the bad, and the not so ugly? *Canadian Field Naturalist*, *129*, 331–341. `https://doi.org/10.22621/cfn.v129i4.1755`

Conferences

Invited talks

Presentations

1. **Newediuk,** L., Melvin, Z., Thomas, L., & Wal, E. V. (2026). When blood is not thicker than water: Social bonds increase use of risky habitat more than kinship. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Toronto, Ontario.
2. Elizabeth Patterson, M. J., Levi **Newediuk.** (2026). Epigenomics as a tool for assessing well-being: A case study in zoo-housed polar bears. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Toronto, Ontario.
3. **Newediuk,** L., Richardson, E. S., Biddlecombe, B. A., Kassim, H., Nicholas Lunn, L. K. ad, Rivkin, L. R., Salama, O. E., Schmidt, C., Jones, M. J., & Garroway, C. J. (2025). Climate change, age acceleration, and the erosion of fitness in polar bears. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Sherbrooke, Quebec.
4. **Newediuk,** L., Schmidt, C., & Garroway, C. J. (2023). Movement behaviour links genetic diversity to the environment. In *Canadian Society for Ecology and Evolution Annual Meeting (Oral presentation)*. Winnipeg, Manitoba.

Symposium organization

Other research outputs

Media coverage

Professional & academic service

Committee roles (Leadership)

Committee roles (Participation)

Other service

Grants & research funding

1903	Nobel Prize in Physics	*Awarded for her work on radioactivity with Pierre Curie and Henri Becquerel*
1911	Nobel Prize in Chemistry	*Awarded for the discovery of radium and polonium*